



Contents lists available at ScienceDirect

Clinical Nutrition ESPEN

journal homepage: <http://www.clinicalnutritionespen.com>

Original article

Impact of sarcopenia and muscle strength on postoperative complication risk following pancreatic resection



Sabrina Wielsøe^a, Anna Sundberg^a, Thomas S. Kristensen^b, Jan Christensen^c,
Martin Sillesen^d, Carsten P. Hansen^d, Stefan K. Burgdorf^d, Bente K. Pedersen^a,
Charlotte Suetta^{e,f}, Jesper F. Christensen^{a,1}, Casper Simonsen^{a,*,1}

^a Centre for Physical Activity Research, Rigshospitalet, University of Copenhagen, Denmark^b Department of Radiology, Copenhagen University Hospital, Rigshospitalet, Copenhagen, Denmark^c Department of Occupational Therapy and Physiotherapy, Copenhagen University Hospital, Rigshospitalet, Copenhagen, Denmark^d Department of Organ Surgery and Transplantation, Copenhagen University Hospital, Rigshospitalet, Copenhagen, Denmark^e Department of Clinical Medicine, Faculty of Health, University of Copenhagen, Copenhagen, Denmark^f Department of Geriatric and Palliative Medicine, Copenhagen University Hospital, Bispebjerg and Frederiksberg, Copenhagen, Denmark

ARTICLE INFO

Article history:

Received 2 September 2024

Accepted 7 October 2024

Keywords:

Sarcopenia

Muscle strength

Muscle mass

Postoperative complications

Cancer

Pancreas

SUMMARY

Background and aims: The association between sarcopenia and postoperative complications has been widely reported in patients with cancer. Yet, the lack of standardized population-specific diagnostic cut-off points and assessments of muscle strength is hampering prospective clinical utilization. Therefore, we aimed to examine the impact of sarcopenia, defined by both regional and international cut-off points, along with various methods of measuring skeletal muscle and muscle strength, on the risk of postoperative complications following pancreatic resection.

Methods: The present prospective observational study enrolled patients scheduled for pancreatic resection. Body composition was assessed by DXA and CT prior to surgery. We applied the algorithm and cut-off points suggested by the European Working Group on Sarcopenia in Older People (EWGSOP) as well as cut-off points from a Danish normative reference population to classify patients as sarcopenic. Physical performance was assessed by usual gait speed while muscle strength was assessed by handgrip strength, leg extensor power, and 30-s sit-to-stand. Postoperative complications within 30 days following surgery were classified according to the Clavien–Dindo classification and the American College of Surgeons National Surgical Quality Improvement Program. Complications graded ≥ 3 according to Clavien–Dindo were considered major complications.

Results: A total of 134 patients with a mean age of 67 years (SD: 9) were enrolled of whom most underwent pancreaticoduodenectomy (64 %). Using international cut-off points, eight patients (7 %) were classified as sarcopenic using CT scans and sarcopenia was associated with an increased risk of major postoperative complications (RR 2.14 [1.33–3.43]). Using DXA, four patients (3 %) were classified as sarcopenic, all of whom experienced a major complication. With regional cut-off points, 16 patients (13 %) were classified as sarcopenic using CT scans, but sarcopenia was not associated with major complications (RR 1.39 [0.80–2.42]). Nine patients (7 %) were classified as sarcopenic using DXA, but sarcopenia was not associated with major complications (RR 1.15 [0.54–2.48]). Across the different muscle strength assessment methods, handgrip strength consistently demonstrated a stronger association with postoperative complications.

Conclusion: Sarcopenia defined according to the EWGSOP criteria and with international cut-off points is associated with an increased risk of postoperative complications following pancreatic resection. Using regionally based cut-off points, the prevalence of sarcopenia is higher, but it does not confer a higher postoperative complication risk. The use of different muscle strength assessment methods results in vastly different estimates of prevalence of low strength and associations with postoperative outcomes.

© 2024 The Author(s). Published by Elsevier Ltd on behalf of European Society for Clinical Nutrition and Metabolism. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

* Corresponding author.

E-mail address: casper.simonsen@regionh.dk (C. Simonsen).¹ CS and JFC contributed equally to this work.

1. Introduction

The prognostic and predictive role of cancer patients' muscular fitness in surgical oncology has evolved significantly over the last decade [1,2]. Indeed, secondary sarcopenia in patients with cancer has gained strong clinical interest, as a wide range of studies have demonstrated an independent association between sarcopenia and critical endpoints including risk of treatment complications, and long-term prognosis [1–6]. Thus, sarcopenia has become an intriguing entity as a clinically relevant, measurable, and modifiable intermediate endpoint which may be assessed and targeted for optimizing patient outcomes.

However, the assessment of sarcopenia is still not successfully integrated into surgical oncology, largely due to the lack of an operational framework to identify high-risk individuals [1]. For example, prevalence estimates of sarcopenia ranging from 19 % to 65 % have been reported in similar surgical populations, attributable to differences in assessment methods and cut-off points [2]. To this end, several high-quality studies have now adopted the diagnostic criteria of the European Working Group on Sarcopenia in Older People (EWGSOP) [7], where sarcopenia originally was defined as low muscle mass in combination with low muscle strength or low physical performance, demonstrating more consistent and higher risk estimates [1]. While the EWGSOP definition is generally well-accepted, its specific operationalization remains debatable. Specifically, in EWGSOP's consensus statement, international cut-off points for European populations are proposed, but it also states that regional cut-off points should be applied when available [8]. Moreover, gold standard methods like computed tomography (CT) and dual x-ray absorptiometry (DXA)-scan are considered equally applicable, but generally display poor levels of agreement [9]. Finally, while the traditionally used gait speed and handgrip strength are simple and low-cost options, other physical performance and muscle strength tests are readily available and recommended [8].

We therefore conducted the present prospective observational cohort study to explore different available assessment methods and standardized cut-off points to identify preoperative sarcopenia in patients undergoing pancreatic resection. Our primary objective was to evaluate the association between preoperative sarcopenia and postoperative complications using different operational approaches to define sarcopenia. These approaches included: international and regional normative reference data derived cut-off points for skeletal muscle, muscle strength, and physical performance; different assessment methods of muscle strength; and skeletal muscle assessed by CT as well as DXA scans.

2. Methods

2.1. Study design

This prospective observational cohort study included patients scheduled for open pancreatic surgery (pancreaticoduodenectomy, total- or distal pancreatectomy) at the Department of Organ Surgery and Transplantation at Rigshospitalet from December 2017 to December 2020. The exclusion criteria were age <18 years, pregnancy, physical or mental disabilities precluding test of muscle function, unable to understand Danish, and preoperative oncological treatment within the last 12 months. Eligible patients were informed of the study and provided written information by a treating physician during an outpatient visit. One to three days before surgery, the participants underwent assessment of body composition, physical function, and muscle strength. The participants received nutritional counseling from a nurse as part of standard care. Information on diagnosis, disease stage, treatment,

comorbidities, and postoperative complications were extracted from the patients' electronic medical records by surgeons and study nurses blinded for sarcopenia status. Sociodemographic factors were assessed by self-report using a questionnaire. All participants signed informed consent before undertaking any study-related procedures. The study was approved by the local scientific ethics committee (H-17003961) and preregistered at [ClinicalTrials.gov](https://www.clinicaltrials.gov) (NCT03347162).

2.2. Sarcopenia definition

We applied the original EWGSOP algorithm [7] to define sarcopenia as prespecified in our protocol. Thus, sarcopenia was defined as low muscle mass in combination with low muscle strength or low physical performance assessed using usual gait speed [7]. Acknowledging that the EWGSOP defined diagnostic criteria for sarcopenia has been updated [8], we included post hoc analyses in which we applied the new diagnostic criteria that defines sarcopenia as low skeletal muscle in combination with low muscle strength. We applied two separate cut-off points for low skeletal muscle mass and muscle strength. First, we applied the international cut-off points proposed by EWGSOP [8]. Second, we applied cut-off points derived from normative reference data [10,11]. As recommended by EWGSOP2 [8], we defined the cut-off points from the normative reference material as 2 standard deviations (SD) below the mean of a healthy, young adult population. Here, we used regional data from the Copenhagen Sarcopenia study [10] and data from USA for the CT scans [11].

2.3. Assessment and cut-off points for body composition, muscle strength, and physical performance

2.3.1. Computed tomography

From preoperative CT scans, skeletal muscle area was measured at the first slice of the third lumbar vertebra where both vertebral transverse processes were visible. The cross-sectional area was quantified using the SliceOmatic software v.4.3 (Tomovision, Montreal, Canada). The Hounsfield (HU) thresholds were -29HU to +150HU for skeletal muscle [12]. The skeletal muscle index (SMI) was defined as skeletal muscle area normalized to height squared (cm^2/m^2). Low skeletal muscle area was defined as an SMI below $45.4 \text{ cm}^2/\text{m}^2$ and $34.4 \text{ cm}^2/\text{m}^2$ for males and females, respectively [11]. We have previously demonstrated that these cut-off points were the cut-off points from healthy populations yielding the highest agreement between CT and DXA for classification of low skeletal muscle [9]. With no regional cut-off points available for CT, the international cut-off points were used in all analyses.

2.3.2. Dual-energy x-ray absorptiometry

A whole-body DXA scan was performed (Lunar iDXA, Lunar Corporation Madison, WI, USA). We defined low skeletal muscle using appendicular lean soft tissue mass (ALM, i.e., lean soft tissue of legs and arms) normalized to height squared. The regional cut-off points for low ALM were $<6.60 \text{ kg}/\text{m}^2$ and $<5.03 \text{ kg}/\text{m}^2$, and the international cut-off points were $<7.0 \text{ kg}/\text{m}^2$ and $<5.5 \text{ kg}/\text{m}^2$ for males and females, respectively [8,10].

2.3.3. Handgrip strength

Maximal isometric handgrip strength of the dominant hand was assessed using a hand-held dynamometer (Saehan DHD-1 Digital Hand Dynamometer, SAEHAN Corporation, South Korea). The patients were seated and instructed to keep the elbow flexed in a 90° angle. Three attempts were provided, and the highest handgrip strength recorded was used as maximal handgrip strength. Low handgrip strength was defined by regional cut-off points as

<36.11 kg and <20.17 kg and by international cut-off points as <27 kg and <16 kg for males and females, respectively [8,10].

2.3.4. 30 second sit-to-stand

The sit-to-stand test was performed using a standardized chair with the seat 45 cm above the ground. From a seated position, the patients were instructed to stand until their hips and knees were fully extended and then immediately sit back down. The patients were provided 30 s to perform as many repetitions as possible. One attempt was provided and low muscle strength based on the sit-to-stand test was defined by regional cut-off points as <16.16 repetitions and <15.10 repetitions for males and females, respectively [10].

2.3.5. Gait speed

Usual gait speed was measured over a 10 m distance. Starting from a standing position, the patients were instructed to walk as if they were going for a walk. With no regional cut-offs available, low usual gait speed was defined by international cut-off point <0.8/s for both males and females in all analyses [8].

2.3.6. Leg extensor power

Leg extensor power was evaluated using a power Rig [13] (Nottingham Power Rig, Queens's Medical Centre Nottingham, NG7 2UH, United Kingdom). The seat was adjusted to allow a knee angle of 15° with the footplate being fully pushed down. The patients were instructed to keep their hands across the chest and to extend the dominant leg as fast and forcefully as possible. The patients performed at least five maximal attempts with additional attempts being provided until the patient did not improve power output in two successive attempts. Low muscle strength based on leg extensor power was defined by regional cut-off points as <227.47 W and <109.65 W for males and females, respectively [10].

2.4. Postoperative outcomes

Postoperative complications were assessed up to 30 days following surgery. The surgical complications were sub-grouped according to the American College of Surgeons National Surgical Quality Improvement Program (ACS-NSQIP) 2018 definitions [14]. Further, the postoperative complications were classified according to the Clavien-Dindo classification [15]. We defined “all complications” as Clavien-Dindo grade ≥ 2 and “major complications” as Clavien-Dindo grade ≥ 3 .

2.5. Statistical analyses

Baseline characteristics of sarcopenic and non-sarcopenic patients were compared using unpaired t-tests and Fisher's exact test for continuous data and categorical data, respectively. The incidence of postoperative complications in the sarcopenic vs. non-sarcopenic group was compared by estimating the risk ratio and the corresponding 95 % confidence interval. The widths of the confidence intervals are not adjusted for multiplicity and as such, these should not be interpreted as causative evidence. We performed no hypothesis testing of differences in risk. In case of zero events in one or both groups, the absolute number of events is presented in both groups but without estimated risk ratios. The agreement between handgrip strength, sit-to-stand, and leg extensor power in classification of individuals with low muscle strength was investigated in the patients with data available for all three assessments. The agreement was assessed using positive and negative percentage agreement as well as using Cohen's kappa interpreted according to Landis and Koch [16]. The linear correlations between the three muscle strength assessment methods were explored using Pearson's correlation. Data are presented as means with SD or

frequencies with percentages unless stated otherwise. The statistical analyses were performed using R statistical software (version 3.6.1 R Foundation for Statistical Computing, Vienna, Austria) and the package metafor [17].

3. Results

This study included 134 patients who underwent pancreatic surgery between December 2017 and December 2020. We excluded 39 patients as they did not undergo assessment of physical fitness prior to surgery and we subsequently excluded 22 patients as they had a non-resectable tumor, had the surgical procedure converted, or due to the pathological examination of the specimen revealing a different diagnosis (Supplementary Fig. 1). The patients had a mean age of 67 (± 9) years, 47 (35 %) were females and the diagnoses included pancreatic ductal adenocarcinoma (60 %), benign intraductal mucinous neoplasm (25 %), neuroendocrine tumors (10 %), and pancreatitis (5 %). Most patients (64 %) underwent pancreaticoduodenectomy i.e., Whipple's procedure. A full overview of patients' characteristics is available in Table 1. The number of patients with data available for the different assessments were as follows: DXA-scans (N = 131), CT-scans (N = 122), usual gait speed (N = 134), handgrip strength (N = 134), sit-to-stand (N = 133), leg extension power (N = 128).

3.1. Sarcopenia based on CT scans

The absolute number of postoperative complications per group as well as the estimated risk ratios are available in Fig. 1. Using the EWG-SOP-algorithm and CT-scans with international sarcopenia cut-off points, eight patients (7 %) were identified as sarcopenic. Six sarcopenic patients (75 %) had a major complication compared with 40 (35 %) in the non-sarcopenic group (RR 2.14 [95 % CI: 1.33–3.43]). Furthermore, sarcopenia was associated with an increased risk of superficial surgical site infection, organ space infection, wound disruption, pneumonia, respiratory failure, cardiac arrest, sepsis, and septic shock.

Using the regional cut-offs, 16 patients were identified as sarcopenic (13 %). Eight patients (50 %) with sarcopenia had a major complication while 38 patients without sarcopenia (36 %) had a major complication (RR 1.39 [95 % CI: 0.80–2.42]). Sarcopenia was associated with an increased risk of wound disruption, pneumonia, respiratory failure, and septic shock.

The risks of postoperative complications associated with sarcopenia when handgrip strength is substituted with by sit to stand and leg extensor power are available in Supplementary Figs. 2 and 3.

3.2. Sarcopenia based on DXA scans

The absolute number of postoperative complications per group as well as the estimated risk ratios are available in Fig. 2. With the international cut-off points, four patients (3 %) were identified as sarcopenic. All sarcopenic patients (100 %) compared with 47 (37 %) of the non-sarcopenic patients had a major complication. Moreover, sarcopenia was associated with an increased risk of death, organ space infection, pneumonia, unplanned intubation, on ventilator 48 h, acute renal failure, cardiac arrest, myocardial infarction, sepsis, and septic shock.

Applying the regional cut-off points, nine patients (7 %) were identified as sarcopenic. Four patients (44 %) with sarcopenia compared with 47 patients (39 %) without sarcopenia had a major complication (RR 1.15 [95 % CI: 0.54–2.48]). Sarcopenia was associated with an increased risk of bleedings requiring transfusion. The risks of postoperative complications associated with sarcopenia

Table 1

Baseline characteristics.

	All patients (n = 134)	Females (n = 47)	Males (n = 87)
Demographics			
Age (years)	67 ± 9	65 ± 10	68 ± 9
Diagnosis			
PDAC	81 (60 %)	27 (57 %)	55 (63 %)
NET	13 (10 %)	3 (6 %)	10 (12 %)
IPMN	33 (25 %)	14 (30 %)	19 (22 %)
Pancreatitis	7 (5 %)	3 (6 %)	4 (5 %)
Charlson comorbidity index (CCI)			
CCI 0	45 (34 %)	13 (28 %)	32 (37 %)
CCI I – II	52 (39 %)	22 (47 %)	30 (35 %)
CCI III – IV	19 (14 %)	4 (9 %)	15 (17 %)
CCI > IV	10 (8 %)	4 (9 %)	6 (7 %)
ASA-score			
ASA I – II	62 (46 %)	23 (49 %)	39 (45 %)
ASA ≥ III	65 (49 %)	20 (43 %)	45 (52 %)
Smoking habits			
Current	21 (16 %)	9 (19 %)	12 (14 %)
Never	48 (36 %)	19 (40 %)	29 (33 %)
Former	57 (43 %)	17 (36 %)	40 (46 %)
Alcohol intake			
Females >7, males >14 units/week	34 (25 %)	12 (26 %)	22 (25 %)
Females <7, males <14 units/week	90 (67 %)	32 (68 %)	58 (67 %)
Surgery type			
Whipple (pancreaticoduodenectomy)	86 (64 %)	32 (68 %)	54 (62 %)
Distal pancreatectomy	27 (20 %)	7 (15 %)	20 (23 %)
Total pancreatectomy	17 (13 %)	6 (13 %)	11 (13 %)
Robot assisted distal pancreatectomy	4 (3 %)	2 (4 %)	2 (2 %)
Body composition			
Body weight (kg)	76.8 ± 15.6	67.9 ± 13.1	81.4 ± 14.8
BMI	25.4 ± 4.5	24.5 ± 4.8	25.9 ± 4.3
Underweight <18.5 kg/m ²	5 (4 %)	4 (9 %)	1 (1.1 %)
Normal 18.6–24.9 kg/m ²	65 (49 %)	25 (53 %)	40 (46 %)
Overweight 25.0–29.9 kg/m ²	48 (36 %)	12 (26 %)	36 (41 %)
Obese >30 kg/m ²	16 (12 %)	6 (13 %)	10 (12 %)
CT-scan at L3			
SMI (cm ² /m ²)	45.2 ± 8.8	39.2 ± 6.6	48.3 ± 8.2
SMA (cm ²)	136.6 ± 32.3	107.8 ± 19.4	151.7 ± 27.0
DXA-scan			
Total lean soft tissue (kg)	49.2 ± 8.9	40.1 ± 5.0	53.8 ± 6.6
App. lean soft tissue/height (kg/m ²)	7.1 ± 1.2	6.2 ± 1.0	7.6 ± 1.1
Total adipose tissue (kg)	24.9 ± 9.9	25.5 ± 9.6	24.6 ± 10.0
Fat percentage (%)	32.6 ± 8.2	37.5 ± 7.9	30.2 ± 7.3
Physical function			
Usual gait speed (m/s)	1.26 ± 0.26	1.27 ± 0.28	1.26 ± 0.24
Handgrip strength (kg)	35.4 ± 9.5	26.8 ± 5.4	39.9 ± 7.9
Leg extension power (W/kg)	2.31 ± 0.90	1.86 ± 0.74	2.57 ± 0.89
Sit-to-stand (rep.)	16 ± 5	16 ± 6	16 ± 5

PDAC, Pancreatic ductal adenocarcinoma; NET, Pancreatic neuroendocrine tumor; IPMN, Intraductal papillary mucinous neoplasms; ASA-score, American Society of Anesthesiologists' classification of Physical Health; SMI, Skeletal Muscle Index; Rep, repetitions; SMA, Skeletal Muscle Area; App. Lean soft tissue, appendicular lean soft tissue. Values are mean (SD) or frequency (%).

when handgrip strength is substituted with sit-to-stand and leg extensor power are available in [Supplementary Figs. 4–7](#).

3.3. Agreement between assessment methods

The prevalence of low skeletal muscle, low muscle strength, and physical performance using regional and international cut-off in men and women is presented in [Table 2](#). Using the regional cut-off points, the prevalence of low upper body muscle strength assessed as handgrip strength (22 %) was lower than the prevalence of low lower body muscle strength assessed via sit-to-stand (59 %) and leg extensor power (52 %). Moreover, the prevalence of low muscle strength was higher than the prevalence of low ALM (15 %) using regional cut-off points. The agreement between handgrip strength, sit-to-stand, and leg extensor power using the regional cut-off points is visualized in [Fig. 3](#). For handgrip strength and leg extensor power, Cohen's Kappa was 0.32 (i.e., indicating fair

agreement) while the positive and negative agreements were 93 % and 56 %, respectively. Cohen's kappa for handgrip strength and sit-to-stand was 0.20 (i.e., indicating slight agreement), with positive and negative agreements of 83 % and 48 %, respectively. Finally, Cohen's kappa for leg extensor power and sit-to-stand was 0.31 with a positive agreement of 73 % and a negative agreement of 59 %.

Low handgrip strength was associated with an increased risk of major postoperative complication (RR 1.72 [95 % CI: 1.15–2.56]) while low sit to stand and leg extensor power were not ([Fig. 4](#)). Similarly, across most of the specific postoperative complications, the estimated risk ratio was higher for low handgrip strength than for sit-to-stand and leg extensor power.

3.4. Post hoc analyses

The post hoc analyses assessing the association between sarcopenia assessed using EWGSOP2 criteria (i.e., low skeletal muscle

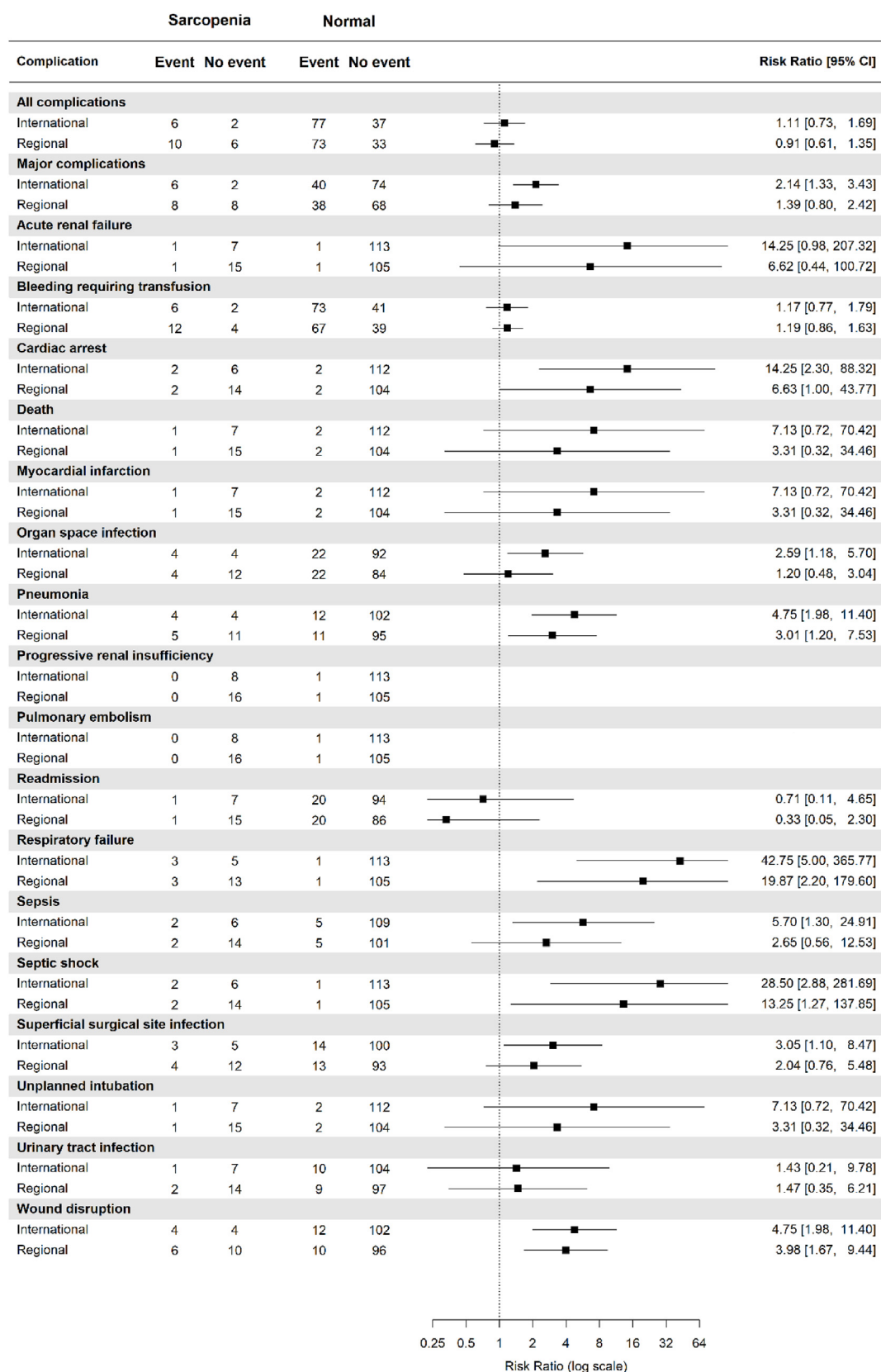


Fig. 1. Risk of postoperative complications by preoperative sarcopenia with skeletal muscle assessed using CT scans. International: International cut-off points for CT, usual gait speed, and hand grip strength. Regional: International cut-off points for CT and usual gait speed, and regional cut-off points for hand grip strength.

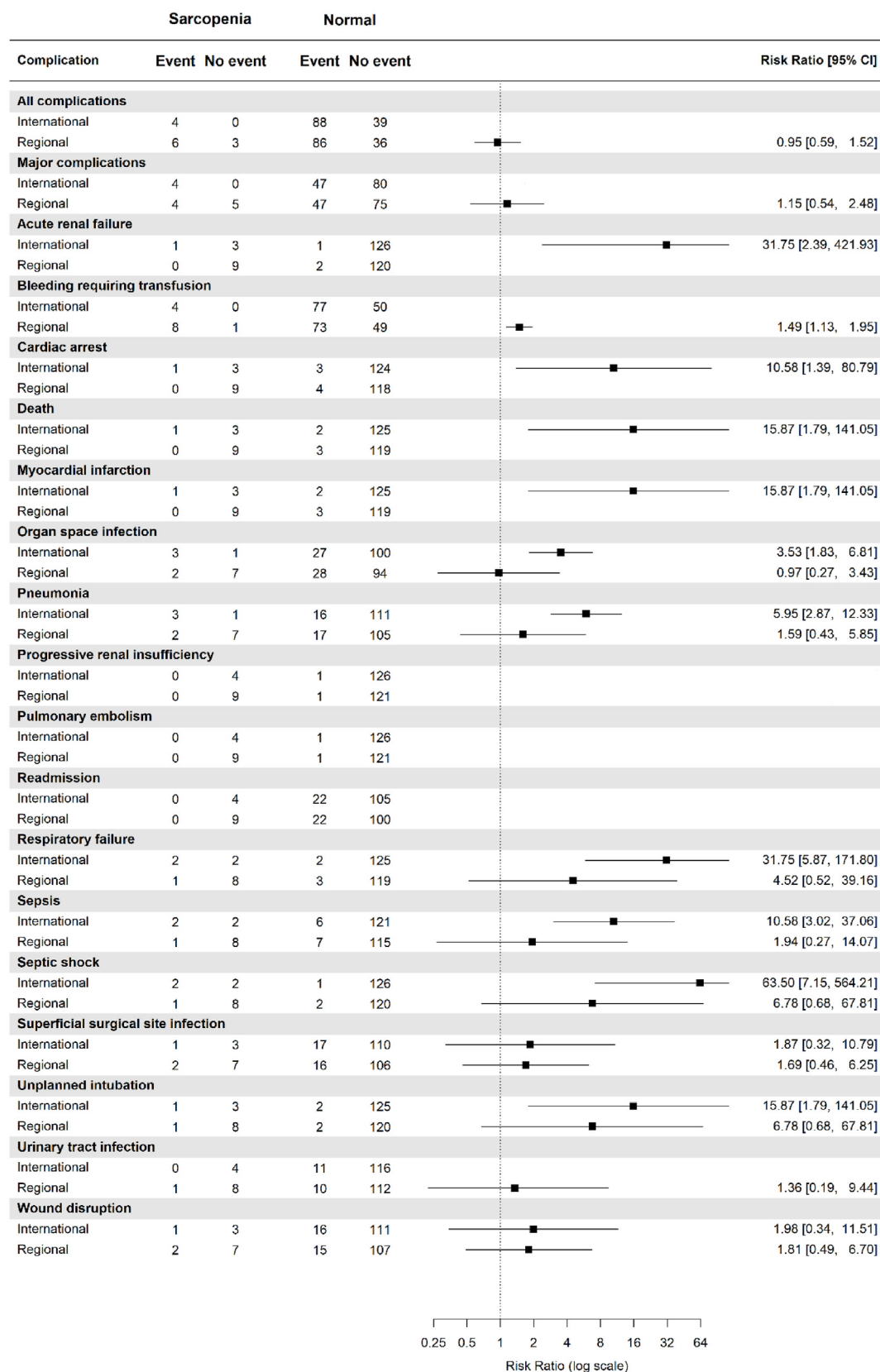


Fig. 2. Risk of postoperative complications by preoperative sarcopenia with skeletal muscle assessed using DXA scans. International: International cut-off points for DXA, usual gait speed, and hand grip strength. Regional: Regional cut-off points for DXA and hand grip strength and international cut-off points for usual gait speed.

Table 2

Prevalence of low skeletal muscle, low muscle strength, and low physical performance using regional and international cut-off points. Numbers are frequencies (%). ALM: appendicular lean soft tissue mass, CT: computed tomography, DXA: dual-energy x-ray absorptiometry, NA: no patients with missing assessment, Reps: repetitions, SMI: skeletal muscle index, W: watt.

	Regional cut-off points			International cut-off points		
	Female (n = 47)	Male (n = 87)	All patients (n = 134)	Female (n = 47)	Male (n = 87)	All patients (n = 134)
ALM DXA	5.03 kg/m²	6.60 kg/m²		5.3 kg/m²	7.0 kg/m²	
Low	5 (11)	15 (17)	20 (15)	7 (15)	25 (29)	32 (24)
Normal	39 (83)	72 (83)	111 (83)	37 (79)	62 (71)	99 (74)
Missing	3 (6)	NA	3 (2)	3 (6)	NA	3 (2)
SMI CT				34.4 cm²/m²	45.4 cm²/m²	
Low	-	-	-	10 (21)	34 (39)	44 (33)
Normal	-	-	-	32 (68)	46 (53)	78 (58)
Missing	-	-	-	5 (11)	7 (8)	12 (9)
Usual gait speed				0.8 m/s	0.8 m/s	
Low	-	-	-	3 (6)	3 (3)	6 (4)
Normal	-	-	-	44 (94)	84 (97)	128 (96)
Missing	-	-	-	NA	NA	NA
Handgrip strength	20.17 kg	36.11 kg		16.0 kg	27.0 kg	
Low	5 (11)	25 (29)	30 (22)	1 (2)	7 (8)	8 (6)
Normal	42 (89)	62 (71)	104 (78)	46 (98)	80 (92)	126 (94)
Missing	NA	NA	NA	NA	NA	NA
Leg extensor power	109.65 W	227.47 W				
Low	21 (45)	49 (56)	70 (52)	-	-	-
Normal	26 (55)	32 (37)	58 (43)	-	-	-
Missing	NA	6 (7)	6 (4)	-	-	-
30 s sit to stand	15.10 reps	16.16 reps				
Low	23 (49)	56 (64)	79 (59)	-	-	-
Normal	23 (49)	31 (36)	54 (40)	-	-	-
Missing	1 (2)	NA	1 (1)	-	-	-

combined with low muscle strength) are available in [Supplementary Figs. 8–17](#). Analyses including patients undergoing pancreaticoduodenectomy only are available in [Supplementary Figs. 18 and 19](#). Analyses including patients with PDAC only are available in [Supplementary Figs. 20 and 21](#). The association between AJCC cancer stage and the prevalence of sarcopenia is provided in [Supplementary Table 1](#).

4. Discussion

Translating an observational, mostly retrospective, evidence base into an operational framework for prospective clinical application constitutes an instrumental challenge for integrating the assessment and management of cancer-related sarcopenia in surgical oncology. Therefore, we aimed to investigate whether sarcopenia assessed using predefined diagnostic criteria would identify individuals with high risk of adverse postoperative outcomes following pancreatic resection. Our main finding was that sarcopenia, based on international cut-off points and using handgrip strength to assess muscle strength, was associated with considerable increased risk of major postoperative complications as well as several of the NSQIP complications. Nevertheless, the prevalence of sarcopenia defined this way was low. Using the regional normative reference data, the prevalence of sarcopenia was higher and the estimated risk ratios were generally lower, but the confidence intervals were largely overlapping the estimates from the international cut-offs. However, it should be noted that nearly 40 % of the patients had a major complication. As such, relying solely on sarcopenia as a risk factor may not effectively encompass the majority of patients who experience major complications. Finally, we demonstrated that the different methods to assess muscle strength displayed poor agreement in classification of patients with low muscle strength. This resulted in substantial differences in the prevalence of low muscle strength, and in the strength of

associations with adverse postoperative outcomes. Altogether, our findings warrant further investigation in larger prospective studies.

4.1. Assessment of sarcopenia, muscle strength, and muscle mass

To date, most studies exploring the prognostic impact of skeletal muscle mass and muscle function in patients undergoing surgery have focused on the prognostic impact of low skeletal muscle mass alone [1,2]. More recently, muscle function and physical performance, primarily assessed using handgrip strength and gait speed, have been included in the assessment of sarcopenia, as recommended by several working groups [8,18,19]. We extend these prior studies by providing a comprehensive assessment of muscular fitness including several assessments of muscle strength, and muscle mass in patients with pancreatic cancer undergoing surgery. Moreover, to our knowledge, the present study is the first to compare cut-off points derived from international guidelines vs. regional normative reference data. The prevalence of sarcopenia in our study was low compared to previous reports (19–65 %) [2]. It should, however, be noted that the prior studies focused on muscle mass alone. Nevertheless, the low prevalence of sarcopenia in this cohort was primarily due to few patients presenting with low skeletal muscle mass as more than half of the participants had low muscle strength or physical performance in tests of lower extremity function. These findings are in line with a previous study that included patients with bladder cancer [20] and imply that loss of physical performance and muscle function may precede a loss of skeletal muscle mass in patients with cancer. Interestingly, handgrip strength is the most widely used method to assess muscle strength, yet we found that the prevalence of low handgrip strength was remarkably lower than the prevalence of low lower-extremity muscle function. This is notable, given that the regional cut-off points for both upper and lower extremity muscle strength originate from the same Danish cohort [10]. It could be speculated

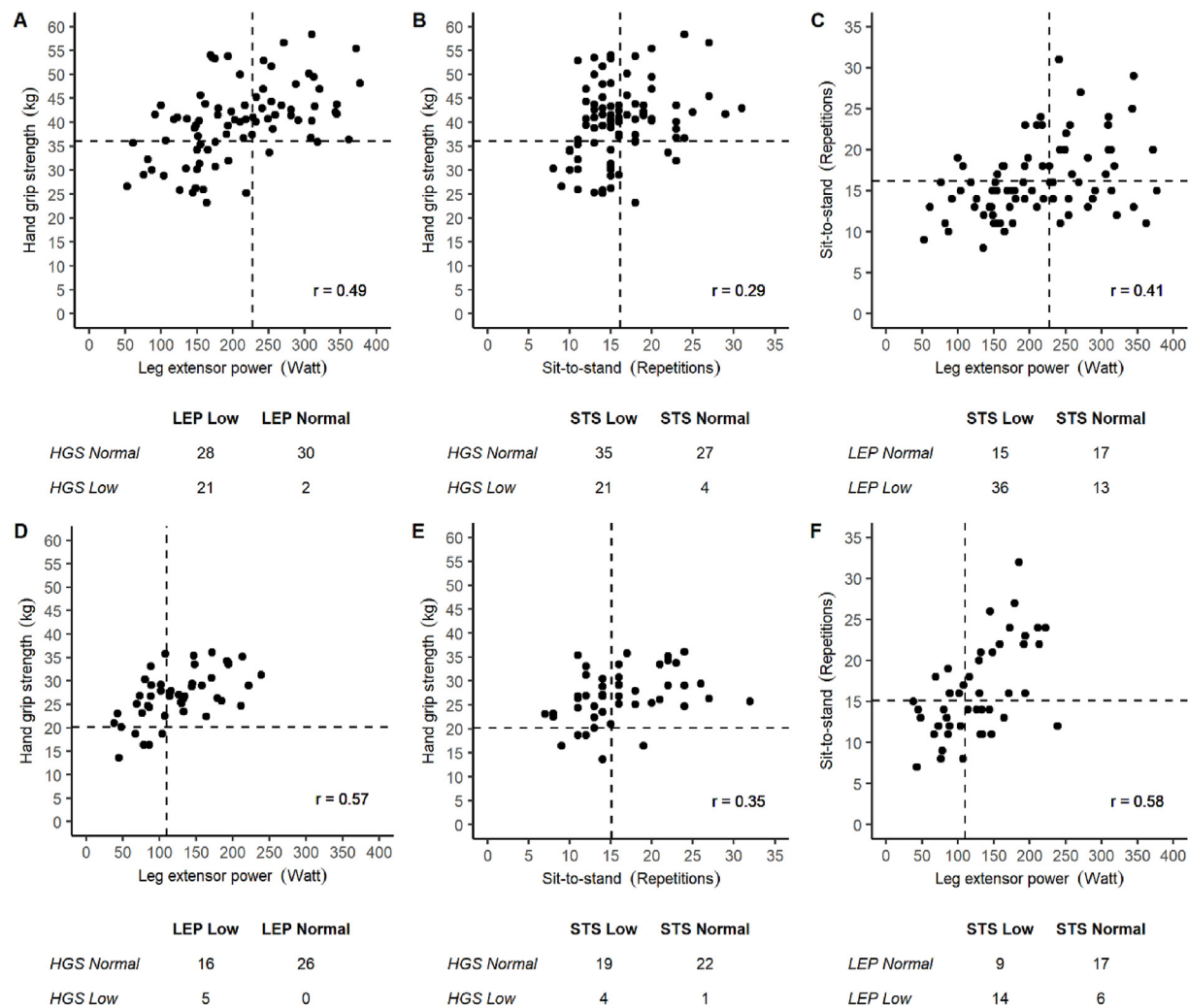


Fig. 3. Scatter plots of muscle strength assessed using hand grip strength, sit-to-stand, and leg extensor power. Vertical and horizontal lines represent the regional cut-off values for the specific muscle strength assessment method. Fig. 3A–C: data for males. Fig. 3D–F: data for females. The 2 × 2 tables represent the agreement between the two muscle strength assessment methods presented in the Figure above. HGS: handgrip strength, LEP: leg extensor power, STS: sit to stand.

that handgrip strength is less sensitive to change, compared to more complex tests of muscular fitness such as tests of gait speed, sit-to-stand or muscle power that require a combination of balance, coordination, and strength. Despite this, handgrip strength showed a stronger association with risk of postoperative complication than the tests of lower extremity muscle function. It should, however, be noted that for the test of lower extremity muscle strength, we used only regional cut-off points. Most patients with pancreatic cancer are older adults, and the prevalence of cachexia among patients with pancreatic cancer is high. Thus, differential effects of ageing and cachexia might help explain these findings. Approximately 50 % of the patients undergoing Whipple's procedure had a stent placed in the bile ducts preoperatively, but cholangitis was rare. Similarly, the proportion of patients with a gastrointestinal obstruction was very low. Comparing CT to DXA, the prevalence of low skeletal muscle was lower with DXA than with CT and we have previously demonstrated that the agreement between CT and DXA for classification of low skeletal muscle is poor [9]. Despite the differences, sarcopenia assessed both using CT and DXA was associated with increased risk of postoperative complications. Future studies should continue to investigate which skeletal muscle and muscle strength assessments methods that show the strongest association with postoperative outcomes.

4.2. Prospective versus retrospective studies on sarcopenia and risk of adverse postoperative outcomes

Most studies investigating impact of body composition on postoperative complications are retrospective and use only diagnostic CT-scans [1,2]. This approach allows for inclusion of consecutive patients and in turn lowers the risk of selection bias. However, due to the retrospective nature of these studies, they are not preregistered with prespecified established cut-off points. Instead, it is common practice to test several different ways to subcategorize patients into sarcopenic and non-sarcopenic sub-groups. This is frequently done using data-specific analytic approaches, such as lowest quartile, median split, or optimal stratification [1,21], resulting in an overwhelming number of suggested cut-off points none of which are applicable in clinical practice without validation. The diverse methodologies and the derived inconsistent findings highlight the importance of a clear uniform agreement on the assessment of sarcopenia and to keep this discrepancy in mind when interpreting findings from different studies. To this end, we undertook a prospective observational cohort study testing pre-defined sarcopenia criteria using cut-off points from a regional normative population of healthy young adults and international consensus cut-off points. While we intended to enroll all eligible

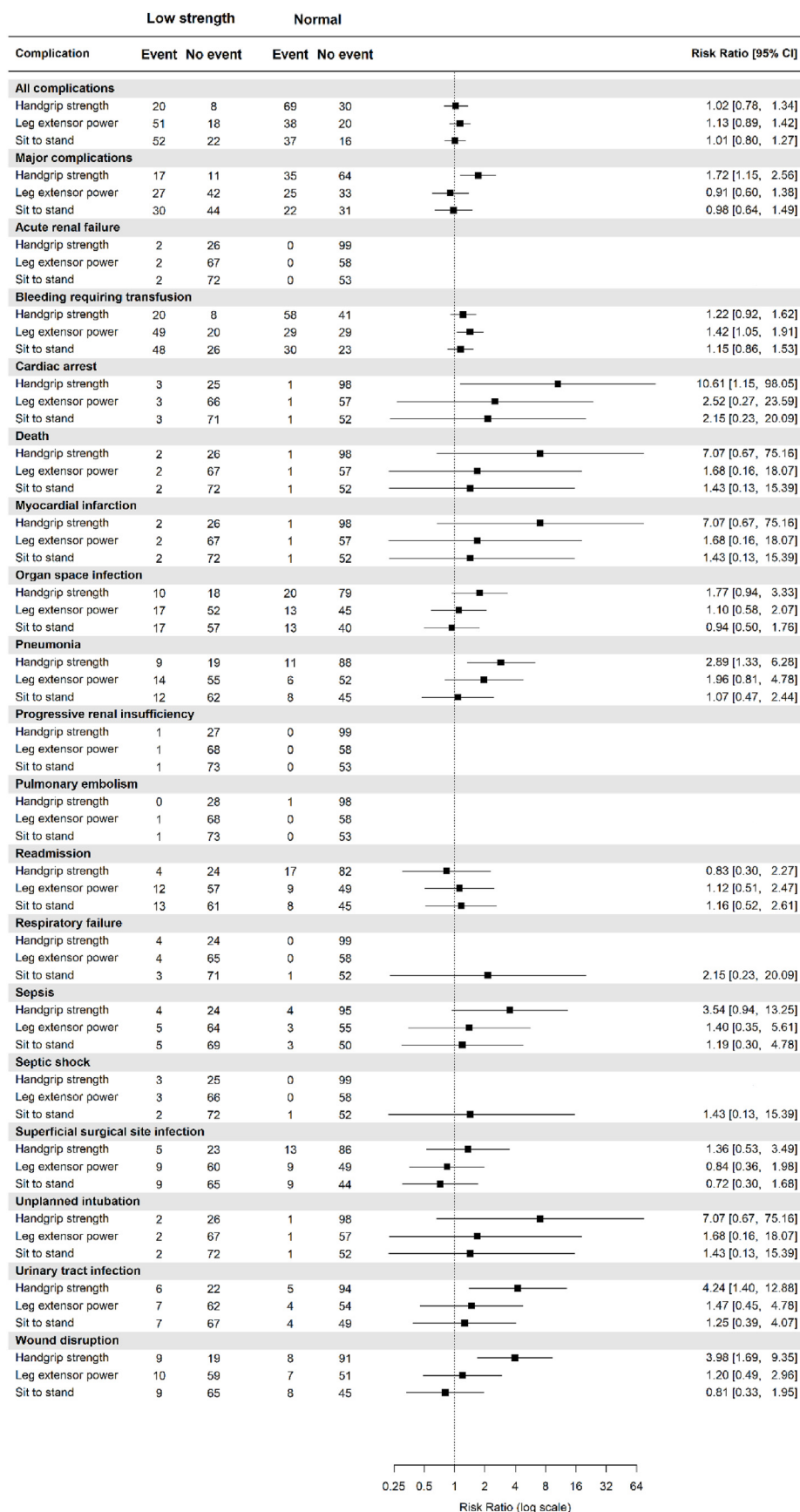


Fig. 4. Risk of postoperative complication associated with low muscle strength assessed by muscle strength, sit to stand test and leg extensor lower. Regional cut-off points from the same population are applied for all assessments of muscle strength. Only participants with data available from all three assessments are included.

consecutive patients during the study period, we cannot discard, that some patients, possibly the most frail/vulnerable ones, declined participation, as the study required additional procedures/tests. As such, we cannot rule out that recruitment of the most well-functioning patients could be one reason for the low prevalence of sarcopenia in the present study. Yet, the prevalence of low muscle strength, especially of the lower extremity, was high within our population (>50 %), and the high overall rate of postoperative complications (69 %) suggests that high-risk patients were indeed included. To overcome potential selection bias in future prospective studies, assessment of physical performance and muscle function most likely need be included as an integrated part of standard preoperative evaluations.

4.3. Strengths and limitations

A strength of our study is that it is prospectively registered, and the prospective design enabled the assessment of muscle mass and muscle strength using various methods. Our study has some important limitations. First, the cut-off points used for low skeletal muscle by DXA and CT originate from different populations. We are, however, not aware of any studies providing cut-off points for sarcopenia using DXA and CT from the same population. Second, we sub-grouped the postoperative complications according to NSQIP to investigate whether sarcopenia increases the risk of different sub-types of postoperative complications. Yet, some of the postoperative complications within NSQIP were rare in our cohort and our ability to assess the association between sarcopenia and the risk of these complications was therefore limited. This precluded estimating risk ratios for several complications as there was zero events in either the sarcopenic or non-sarcopenic group. Moreover, due to the low number of complications in the sarcopenia group, the true risk associated with sarcopenia is likely to vary from this study's point estimates. Our findings warrant further investigation in larger prospective studies.

5. Conclusion

Sarcopenia defined by international cut-off points was a strongly associated with postoperative complications following pancreatic resection. Sarcopenia defined by regional cut-off points resulted in a higher prevalence of sarcopenia with lower risk estimates but largely overlapping the confidence intervals of the international estimates. Different methods to assess muscle strength and skeletal muscle mass demonstrate poor agreement and provide differing prognostic information. Future studies are warranted to further develop and validate an operational framework to properly assess and manage sarcopenia within surgical oncology.

Author contribution

Sabrina Wielsøe: Data curation; investigation; formal analysis; project administration; writing – original draft; writing – review & editing. Anna Sundberg: Data curation; project administration; writing – review & editing. Thomas S. Kristensen: Data curation; investigation; writing – review & editing. Jan Christensen: Data curation; investigation; writing – review & editing. Martin Sillesen: Data curation; investigation; writing – review & editing. Carsten P. Hansen: Data curation; investigation; writing – review & editing. Stefan K. Burgdorf: Data curation; investigation; writing – review & editing. Bente Klarlund Pedersen: writing – review & editing; resources; supervision. Charlotte Suetta: Conceptualization; writing – review & editing. Jesper Frank Christensen: Conceptualization; investigation; writing – original draft; writing – review & editing; funding acquisition; supervision. Casper

Simonsen: Conceptualization; formal analysis; investigation; data curation; writing – original draft; writing – review & editing; visualization; supervision.

Funding

This study was supported by grants from the Danish Cancer Society (grant number R231-A13997), the Svend Andersen Foundation, and the Lundbeck Foundation (grant number R238-2016-2821). The Centre for Physical Activity Research (CFAS) is supported by TrygFonden (grant numbers ID 101390, ID 20045, and ID 125132). The funding sources had no role in the design and conduct of the study; collection of data, analysis, and interpretation; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgement

The authors thank the physiotherapists; Magnus Nygaard Bech, Peter Hendeliowitz, Andreas Ubberup Larsen and Charlotte Grønset from the Department of Occupational Therapy and Physiotherapy Copenhagen University Hospital, Rigshospitalet, Denmark for their assistance with test assessment. Also, the authors wish to thank Anja Kort, Mette Barrit, and Ea Bjunes, Digestive Disease Center, Bispebjerg Hospital, for their invaluable assistance in registration of postoperative complications.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnesp.2024.10.003>.

References

- [1] Simonsen C, de Heer P, Bjerre ED, Suetta C, Hojman P, Pedersen BK, et al. Sarcopenia and postoperative complication risk in gastrointestinal surgical oncology. *Ann Surg* 2018;268:58–69. <https://doi.org/10.1097/SLA.0000000000002679>.
- [2] Bundred J, Kamarajah SK, Roberts KJ. Body composition assessment and sarcopenia in patients with pancreatic cancer: a systematic review and meta-analysis. *HPB* 2019;21:1603–12. <https://doi.org/10.1016/j.hpb.2019.05.018>.
- [3] Joglekar S, Asghar A, Mott SL, Johnson BE, Button AM, Clark E. Sarcopenia is an independent predictor of complications following pancreatectomy for adenocarcinoma. *J Surg Oncol* 2015;111:771–5. <https://doi.org/10.1002/jso.23862>.
- [4] Wang H, Yang R, Xu J, Fang K, Abdelrahim M, Chang L. Sarcopenia as a predictor of postoperative risk of complications, mortality and length of stay following gastrointestinal oncological surgery. *Ann R Coll Surg Engl* 2021. <https://doi.org/10.1308/RCSANN.2021.0082/ASSET/IMAGES/SMALL/RCSANN.2021.0082.05.GIF>.
- [5] Sugimoto M, Farnell MB, Nagorney DM, Kendrick ML, Truty MJ, Smoot RL, et al. Decreased skeletal muscle volume is a predictive factor for poorer survival in patients undergoing surgical resection for pancreatic ductal adenocarcinoma. *J Gastrointest Surg* 2018;22:831–9. <https://doi.org/10.1007/s11605-018-3695-z>.
- [6] Amini N, Spolverato G, Gupta R, Margonis GA, Kim Y, Wagner D, et al. Impact total psoas volume on short- and long-term outcomes in patients undergoing curative resection for pancreatic adenocarcinoma: a new tool to assess sarcopenia. *J Gastrointest Surg* 2015;19:1593. <https://doi.org/10.1007/s11605-015-2835-Y>.
- [7] Cruz-Jentoft AJ, Pierre Baeyens J, Bauer JM, Boirie Y, Cederholm T, Landi F, Sarcopenia, et al. European consensus on definition and diagnosis report of the European working group on sarcopenia in older people. *Age Ageing* 2010;39:412–23. <https://doi.org/10.1093/ageing/afq034>.
- [8] Cruz-Jentoft AJ, Bahat G, Bauer J, Boirie Y, Bruyère O, Cederholm T, et al. Sarcopenia: revised European consensus on definition and diagnosis. *Age Ageing* 2019;48:16–31. <https://doi.org/10.1093/ageing/afy169>.
- [9] Simonsen C, Kristensen TS, Sundberg A, Wielsøe S, Christensen J, Hansen CP, et al. Assessment of sarcopenia in patients with upper gastrointestinal tumors:

- prevalence and agreement between computed tomography and dual-energy x-ray absorptiometry 2021. <https://doi.org/10.1016/j.clnu.2021.03.022>.
- [10] Suetta C, Haddock B, Alcazar J, Noerst T, Hansen OM, Ludvig H, et al. The copenhagen sarcopenia study: lean mass, strength, power, and physical function in a danish cohort aged 20–93 years. *J Cachexia Sarcopenia Muscle* 2019;10:1316–29. <https://doi.org/10.1002/jcsm.12477>.
 - [11] Derstine BA, Holcombe SA, Ross BE, Wang NC, Su GL, Wang SC. Skeletal muscle cutoff values for sarcopenia diagnosis using T10 to L5 measurements in a healthy US population. *Sci RepoRtS* | 2018;8:11369. <https://doi.org/10.1038/s41598-018-29825-5>.
 - [12] Mitsiopoulos N, Baumgartner RN, Heymsfield SB, Lyons W, Gallagher D, Ross R, et al. Cadaver validation of skeletal muscle measurement by magnetic resonance imaging and computerized. tomography 1998.
 - [13] Skelton DA, Greig CA, Davies JM, Young A. Strength, power and related functional ability of healthy people aged 65–89 years. *Age Ageing* 1994;23: 371–7. <https://doi.org/10.1093/AGEING/23.5.371>.
 - [14] Bilimoria KY, Bentrem DJ, Ko CY, Ritchey J, Stewart AK, Winchester DP, et al. Validation of the 6 th edition AJCC pancreatic cancer staging system report from the national cancer database BACKGROUND. With the development of stage-specific treatments for pancreatic 2007. <https://doi.org/10.1002/cncr.22852>.
 - [15] Dindo D, Demartines N, Clavien PA. Classification of surgical complications: a new proposal with evaluation in a cohort of 6336 patients and results of a survey. *Ann Surg* 2004;240:205–13. <https://doi.org/10.1097/01.SLA.0000133083.54934.AE>.
 - [16] Landis JR, Koch GG. The measurement of observer Agreement for categorical data. *Biometrics* 1977;33:159–74. <https://doi.org/10.2307/2529310>.
 - [17] Viechtbauer W. Conducting meta-analyses in R with the metafor package. *J Stat Softw* 2010;36:1–48. <https://doi.org/10.18637/jss.v036.i03>.
 - [18] Kirk B, Cawthon PM, Arai H, Ávila-Funes JA, Barazzoni R, Bhasin S, et al. The conceptual definition of sarcopenia: delphi consensus from the global leadership initiative in sarcopenia (GLIS). *Age Ageing* 2024;53. <https://doi.org/10.1093/ageing/afae052>.
 - [19] Chen L-K, Woo J, Assantachai P, Auyeung T-W, Chou M-Y, Iijima K, et al. Asian working group for sarcopenia: 2019 consensus update on sarcopenia diagnosis and treatment. *J Am Med Dir Assoc* 2020;21:300–307.e2. <https://doi.org/10.1016/j.jamda.2019.12.012>.
 - [20] Omeland LH, Ammitzbøll G, Dalton O, Suetta C, Pappot H. Muscle mass and physical function in patients with bladder cancer—data from a prematurely terminated prospective cohort study. *Front Rehabil Sci* 2022;6–11. <https://doi.org/10.3389/fresc.2022.942475>.
 - [21] Chan MY, Siu K, Chok H. Sarcopenia in pancreatic cancer-effects on surgical outcomes and chemotherapy. *World J Gastrointest Oncol* 2019;11:527–37. <https://doi.org/10.4251/wjgo.v11.i7.527>.